

Generation Interconnection System Impact Study Report

For Queue Project Q19-1321

Casky 138 kV
December 17, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1321. The Interconnection Customer proposes a 92.4 MW solar photovoltaic facility interconnecting at the Casky 138 kV substation in the New Jersey area. The request was received on October 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 315 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$319,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
OYSTER CREEK-S SAN ANTONIO 138kV	SOUTHGATE-KELLY LAKE 138kV	105.0	259	15.3
FENNER-SOUTH SAN ANTONIO 138kV	SOUTHGATE SUBSTATION-KELLY LAKE 138kV	104.0	249	15.3
FENNER-CALEXICO 138kV line	SOUTHGATE-KELLY LAKE 138kV	103.4	239	15.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
OYSTER CREEK-S SAN ANTONIO 138kV	Reconductor line	\$2,061,000	8% (\$162,000)
FENNER-SOUTH SAN ANTONIO 138kV	Reconductor line	\$717,000	8% (\$57,000)
FENNER-CALEXICO 138kV	Reconductor line	\$1,264,000	8% (\$100,000)
	Total allocated to project		\$319,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.